# Lecture 14: Two-Port Circuits

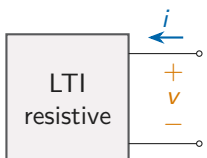
## EE201 – Circuit Theory I

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Middle East Technical University

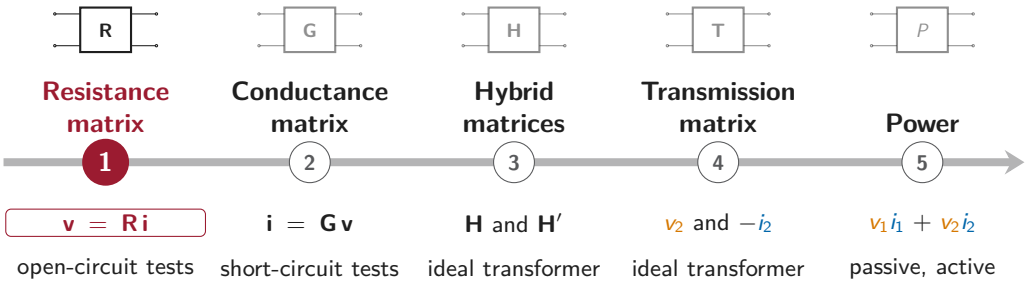
current voltage, node voltage $e$

# Review: one-ports

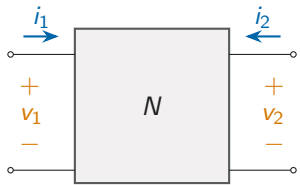



- No independent sources: $v = R_{\text{in}}\, i$
- With independent sources: $v = R_{\text{th}}\, i + v_{\text{th}}$

# Roadmap

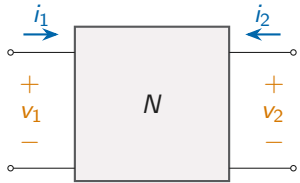

R
G
H
T
P
Resistance matrix
Conductance matrix
Hybrid matrices
Transmission matrix
Power
1
2
3
4
5
v = Ri
i = Gv
H and H′
v2 and −i2
v1i1 + v2i2
open-circuit tests
short-circuit tests
ideal transformer
ideal transformer
passive, active

# Motivation: two ports



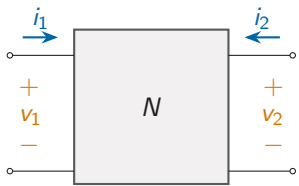

- $N$: LTI resistive, no independent sources
- An input and an output: an amplifier, a transformer, a filter
- One-port: $v = R\,i$ with scalars

# Motivation: two ports



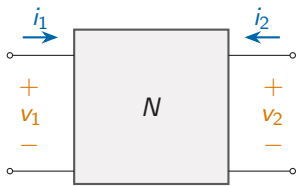

- $N$: LTI resistive, no independent sources
- An input and an output: an amplifier, a transformer, a filter
- One-port: $v = R\,i$ with scalars
- Two-port: $\mathbf{v} = \mathbf{R}\,\mathbf{i}$ with vectors in $\mathbb{R}^2$ and a $2 \times 2$ matrix

# The port variables



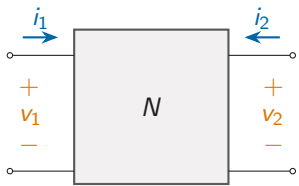

- Each port: the current enters at $+$ and leaves at $-$
- Four variables: $v_1$, $i_1$, $v_2$, $i_2$

# The port variables




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- Four variables: $v_1$, $i_1$, $v_2$, $i_2$
- Two equations: two variables in terms of the other two

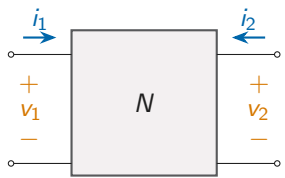
# The port variables




- Each port: the current enters at + and leaves at −
- Four variables: $v_1$, $i_1$, $v_2$, $i_2$
- Two equations: two variables in terms of the other two

| given | find | matrix |
|---|---|---|
| $i_1$, $i_2$ | $v_1$, $v_2$ | resistance **R** |
| $v_1$, $v_2$ | $i_1$, $i_2$ | conductance **G** |
| $i_1$, $v_2$ | $v_1$, $i_2$ | hybrid **H** |
| $v_2$, $-i_2$ | $v_1$, $i_1$ | transmission **T** |

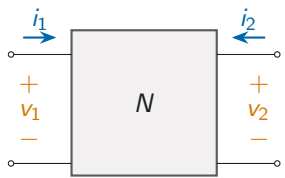
# The resistance matrix




$$v_1 = r_{11}\, i_1 + r_{12}\, i_2$$

$$v_2 = r_{21}\, i_1 + r_{22}\, i_2$$

# The resistance matrix



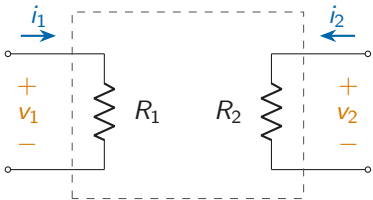

$$v_1 = r_{11}\, i_1 + r_{12}\, i_2$$

$$v_2 = r_{21}\, i_1 + r_{22}\, i_2$$

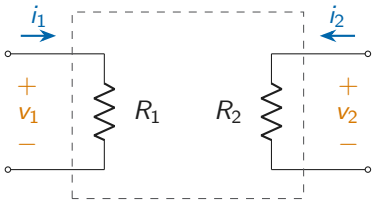
$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \underbrace{\begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix}}_{\mathbf{R}} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

# Example 14.1: two separate resistors

$\mathbf{R} = ?$



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$\mathbf{R} = ?$

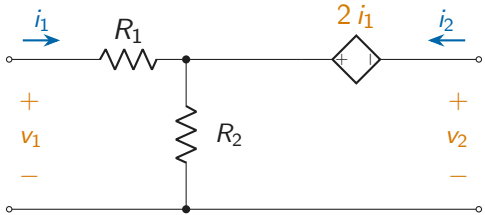
$v_1 = R_1\, i_1 + 0 \cdot i_2$

$v_2 = 0 \cdot i_1 + R_2\, i_2$

$$\mathbf{R} = \begin{bmatrix} R_1 & 0 \\ 0 & R_2 \end{bmatrix}$$

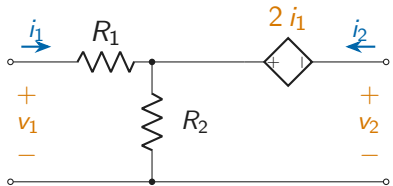
No coupling: $r_{12} = r_{21} = 0$

# Example 14.2: a dependent source inside




$$\mathbf{R} = ?$$

# Example 14.2: KVL

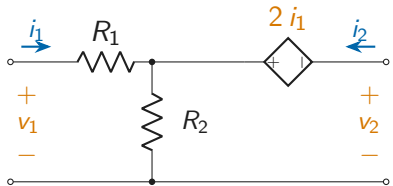



$R_2$ carries $i_1 + i_2$

$v_1 = R_1\, i_1 + R_2(i_1 + i_2)$

$v_2 = -2\, i_1 + R_2(i_1 + i_2)$

# Example 14.2: KVL




$R_2$ carries $i_1 + i_2$

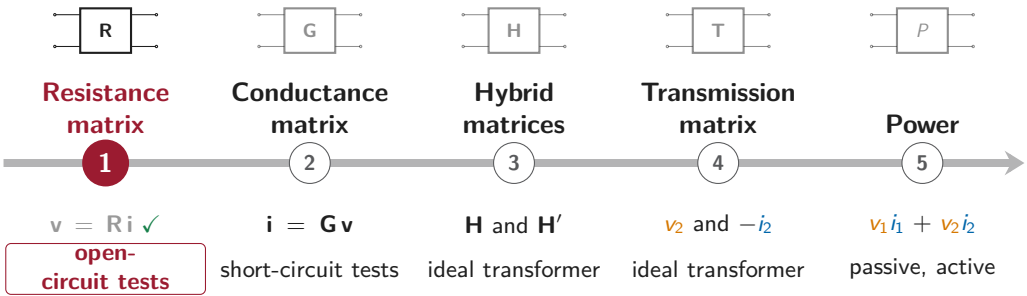
$v_1 = R_1\, i_1 + R_2(i_1 + i_2)$

$v_2 = -2\, i_1 + R_2(i_1 + i_2)$

$$\mathbf{R} = \begin{bmatrix} R_1 + R_2 & R_2 \\ R_2 - 2 & R_2 \end{bmatrix}$$

$r_{12} \neq r_{21}$: the dependent source

# Roadmap


R
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Resistance matrix
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v = Ri ✓
open-circuit tests
i = Gv
short-circuit tests
H and H′
ideal transformer
$v_2$ and $-i_2$
ideal transformer
$v_1 i_1 + v_2 i_2$
passive, active

# The entries from two tests

**Port 2 open:** $i_2 = 0$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix} \begin{bmatrix} i_1 \\ 0 \end{bmatrix} = \begin{bmatrix} r_{11}\, i_1 \\ r_{21}\, i_1 \end{bmatrix} \quad \Rightarrow \quad r_{11} = \frac{v_1}{i_1}, \quad r_{21} = \frac{v_2}{i_1}$$

the first column: $r_{11}$ input resistance, $r_{21}$ transfer resistance

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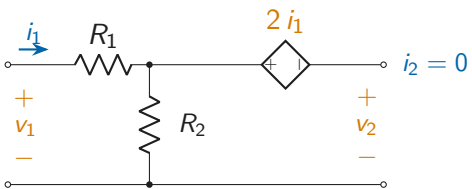
the first column: $r_{11}$ input resistance, $r_{21}$ transfer resistance

**Port 1 open:** $i_1 = 0$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix} \begin{bmatrix} 0 \\ i_2 \end{bmatrix} = \begin{bmatrix} r_{12}\, i_2 \\ r_{22}\, i_2 \end{bmatrix} \quad \Rightarrow \quad r_{12} = \frac{v_1}{i_2}, \quad r_{22} = \frac{v_2}{i_2}$$

the second column: $r_{12}$ transfer resistance, $r_{22}$ input resistance

# Example 14.2 revisited: port 2 open

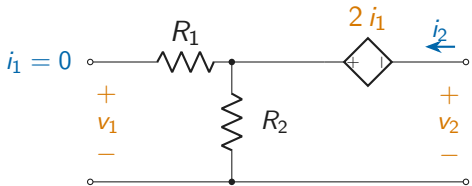



No current through the dependent source

$$v_1 = (R_1 + R_2)\, i_1 \;\Rightarrow\; r_{11} = R_1 + R_2$$

$$v_2 = -2\, i_1 + R_2\, i_1 \;\Rightarrow\; r_{21} = R_2 - 2$$

# Example 14.2 revisited: port 1 open

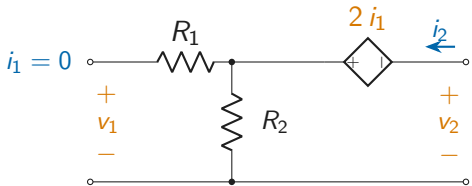



No current in $R_1$; the dependent source is 0

$v_1 = R_2\, i_2 \;\Rightarrow\; r_{12} = R_2$

$v_2 = R_2\, i_2 \;\Rightarrow\; r_{22} = R_2$

# Example 14.2 revisited: port 1 open



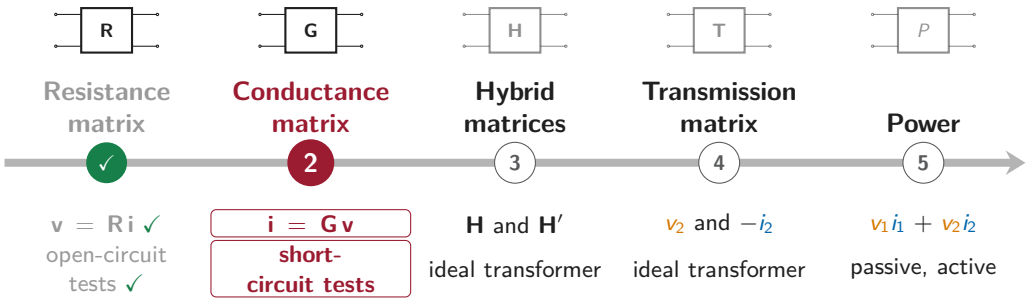

No current in $R_1$; the dependent source is 0

$v_1 = R_2\, i_2 \;\Rightarrow\; r_{12} = R_2$

$v_2 = R_2\, i_2 \;\Rightarrow\; r_{22} = R_2$

The same **R** ✓

# Roadmap


R
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Resistance matrix
Conductance matrix
Hybrid matrices
Transmission matrix
Power
✓
2
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v = Ri ✓
open-circuit tests ✓
i = Gv
short-circuit tests
H and H′
ideal transformer
$v_2$ and $-i_2$
ideal transformer
$v_1 i_1 + v_2 i_2$
passive, active

# The conductance matrix

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \underbrace{\begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}}_{\mathbf{G}} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

# The conductance matrix

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When $\mathbf{R}^{-1}$ exists: $\mathbf{G} = \mathbf{R}^{-1}$

# The entries from two short-circuit tests

**Port 2 shorted:** $v_2 = 0$

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ 0 \end{bmatrix} = \begin{bmatrix} g_{11}\, v_1 \\ g_{21}\, v_1 \end{bmatrix} \quad \Rightarrow \quad g_{11} = \frac{i_1}{v_1}, \quad g_{21} = \frac{i_2}{v_1}$$

the first column: $g_{11}$ input conductance, $g_{21}$ transfer conductance

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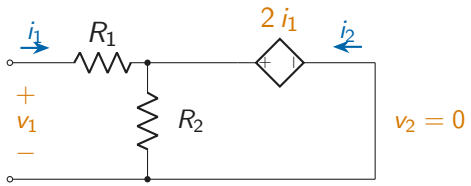
the first column: $g_{11}$ input conductance, $g_{21}$ transfer conductance

**Port 1 shorted:** $v_1 = 0$

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} 0 \\ v_2 \end{bmatrix} = \begin{bmatrix} g_{12}\, v_2 \\ g_{22}\, v_2 \end{bmatrix} \quad \Rightarrow \quad g_{12} = \frac{i_1}{v_2}, \quad g_{22} = \frac{i_2}{v_2}$$

the second column: $g_{12}$ transfer conductance, $g_{22}$ input conductance

# Example 14.3: G for Example 14.2

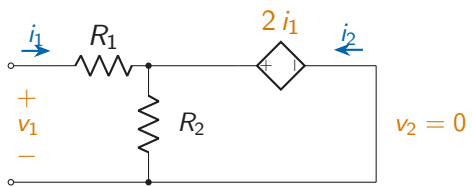



Port 2 shorted: $v_2 = 0$

**Mesh 1:** $-v_1 + R_1\, i_1 + R_2(i_1 + i_2) = 0$

**Mesh 2:** $-2\, i_1 + R_2(i_1 + i_2) = 0$

# Example 14.3: G for Example 14.2




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**Mesh 1:** $-v_1 + R_1\,i_1 + R_2(i_1 + i_2) = 0$

**Mesh 2:** $-2\,i_1 + R_2(i_1 + i_2) = 0$

$$\begin{bmatrix} R_1 + R_2 & R_2 \\ R_2 - 2 & R_2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} v_1$$

The matrix is **R**

# Example 14.3: the result

$$\Delta = \det \mathbf{R} = (R_1 + R_2)\, R_2 - R_2(R_2 - 2) = R_2\,(R_1 + 2)$$

$$g_{11} = \frac{R_2}{\Delta} = \frac{1}{R_1 + 2}, \qquad g_{21} = \frac{2 - R_2}{\Delta}$$

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$$\mathbf{G} = \mathbf{R}^{-1} = \frac{1}{\Delta} \begin{bmatrix} R_2 & -R_2 \\ 2 - R_2 & R_1 + R_2 \end{bmatrix}$$

# Example 14.3: the result

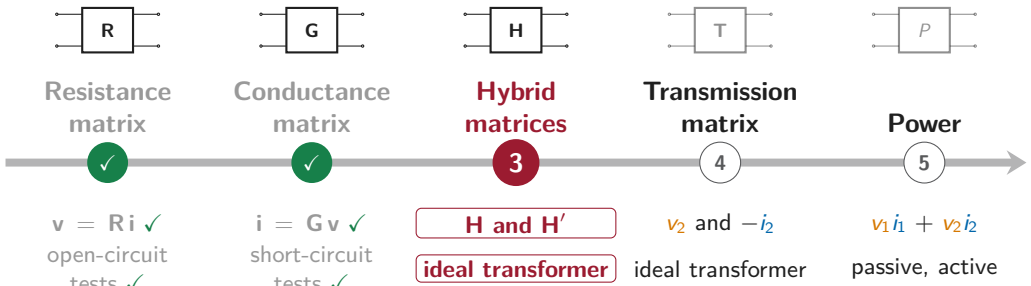
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$$\mathbf{G} = \mathbf{R}^{-1} = \frac{1}{\Delta} \begin{bmatrix} R_2 & -R_2 \\ 2 - R_2 & R_1 + R_2 \end{bmatrix}$$

**R** or **G** need not exist

# Roadmap


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Resistance matrix
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H and H′
ideal transformer
v2 and −i2
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v1 i1 + v2 i2
passive, active

# Hybrid matrices

$$\begin{bmatrix} v_1 \\ i_2 \end{bmatrix} = \underbrace{\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}}_{\mathbf{H}} \begin{bmatrix} i_1 \\ v_2 \end{bmatrix}$$

# Hybrid matrices

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$$\begin{bmatrix} i_1 \\ v_2 \end{bmatrix} = \mathbf{H}' \begin{bmatrix} v_1 \\ i_2 \end{bmatrix}$$

- $\mathbf{H}' = \mathbf{H}^{-1}$ when the inverse exists

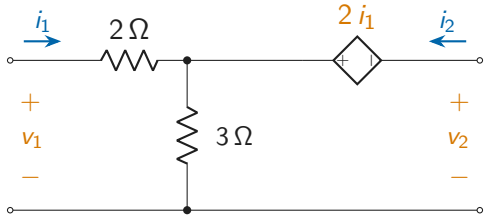
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$$\begin{bmatrix} i_1 \\ v_2 \end{bmatrix} = \mathbf{H}' \begin{bmatrix} v_1 \\ i_2 \end{bmatrix}$$

- $\mathbf{H}' = \mathbf{H}^{-1}$ when the inverse exists
- $h_{11}$, $h_{21}$: port 2 shorted, $v_2 = 0$
- $h_{12}$, $h_{22}$: port 1 open, $i_1 = 0$

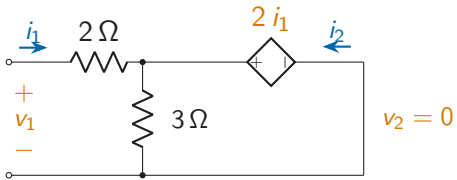
# Example 14.4: H of a circuit




The circuit of Example 14.2 with numbers: $\mathbf{H} = ?$

# Example 14.4: two tests

**Port 2 shorted:** $v_2 = 0$




Mesh 2: $-2\,i_1 + 3(i_1 + i_2) = 0$

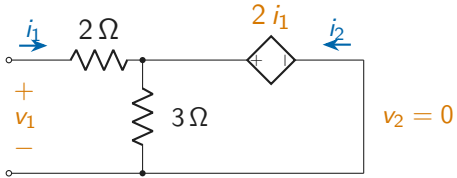
$\Rightarrow \quad i_2 = -\frac{1}{3}\,i_1$

Mesh 1: $v_1 = 2\,i_1 + 3(i_1 + i_2) = 4\,i_1$

$h_{11} = 4\,\Omega, \qquad h_{21} = -\frac{1}{3}$

# Example 14.4: two tests

**Port 2 shorted:** $v_2 = 0$



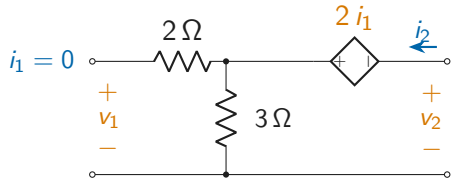

Mesh 2: $-2\,i_1 + 3(i_1 + i_2) = 0$

$\Rightarrow \quad i_2 = -\frac{1}{3}\,i_1$

Mesh 1: $v_1 = 2\,i_1 + 3(i_1 + i_2) = 4\,i_1$

$h_{11} = 4\,\Omega, \qquad h_{21} = -\frac{1}{3}$

**Port 1 open:** $i_1 = 0$




The dependent source is 0: $v_2 = 3\,i_2$

$\Rightarrow \quad i_2 = \frac{1}{3}\,v_2$

$v_1 = 3\,i_2 = v_2$

$h_{12} = 1, \qquad h_{22} = \frac{1}{3}\,\mathrm{S}$

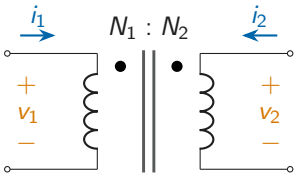
# Example 14.4: the result

$$\mathbf{H} = \begin{bmatrix} 4 & 1 \\ -\frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

- Units: $h_{11}$ in $\Omega$, $h_{22}$ in S, $h_{12}$ and $h_{21}$ without units
- Check with $\mathbf{R} = \begin{bmatrix} 5 & 3 \\ 1 & 3 \end{bmatrix}$ from Example 14.2 ✓

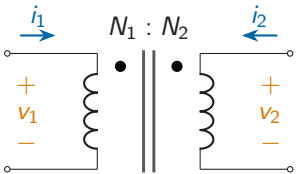
# Example 14.5: the ideal transformer

$\mathbf{H} = ?$ $\mathbf{R} = ?$




$$\frac{v_1}{N_1} = \frac{v_2}{N_2}, \quad N_1\, i_1 + N_2\, i_2 = 0$$

# Example 14.5: the ideal transformer




$$\frac{v_1}{N_1} = \frac{v_2}{N_2}, \qquad N_1\, i_1 + N_2\, i_2 = 0$$

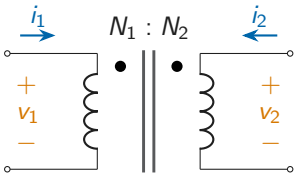
$\mathbf{H} = ?$ $\mathbf{R} = ?$

With $n = N_1/N_2$:

$v_1 = n\, v_2, \qquad i_2 = -n\, i_1$

$$\mathbf{H} = \begin{bmatrix} 0 & n \\ -n & 0 \end{bmatrix}$$

# Example 14.5: the ideal transformer




$$\frac{v_1}{N_1} = \frac{v_2}{N_2}, \qquad N_1\, i_1 + N_2\, i_2 = 0$$

$\mathbf{H} = ?$ $\mathbf{R} = ?$

With $n = N_1/N_2$:

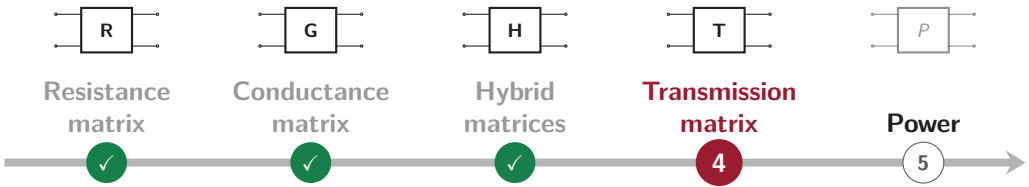
$v_1 = n\, v_2, \qquad i_2 = -n\, i_1$

$$\mathbf{H} = \begin{bmatrix} 0 & n \\ -n & 0 \end{bmatrix}$$

Ports open: $i_1 = i_2 = 0$, but $v_1$, $v_2$ are free

No $\mathbf{R}$, no $\mathbf{G}$

# Roadmap


R
G
H
T
P
Resistance matrix
Conductance matrix
Hybrid matrices
Transmission matrix
Power
✓
✓
✓
4
5
v = Ri ✓
open-circuit tests ✓
i = Gv ✓
short-circuit tests ✓
H and H′ ✓
ideal transformer ✓
v2 and −i2
ideal transformer
v1i1 + v2i2
passive, active

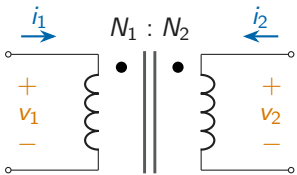
# The transmission matrix

$$\begin{bmatrix} v_1 \\ i_1 \end{bmatrix} = \underbrace{\begin{bmatrix} A & B \\ C & D \end{bmatrix}}_{\mathbf{T}} \begin{bmatrix} v_2 \\ -i_2 \end{bmatrix}$$

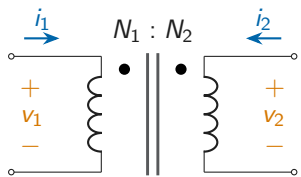
- Also called the chain parameters
- Port 1 in terms of port 2
- $-i_2$: the current that leaves port 2

# Example 14.6: T of the ideal transformer

$$\mathbf{T} = ?$$



# Example 14.6: T of the ideal transformer




$$\mathbf{T} = ?$$

$$v_1 = \frac{N_1}{N_2}\, v_2$$

$$i_1 = -\frac{N_2}{N_1}\, i_2 = \frac{N_2}{N_1}\,(-i_2)$$

$$\mathbf{T} = \begin{bmatrix} N_1/N_2 & 0 \\ 0 & N_2/N_1 \end{bmatrix}$$

# Which test? Open or short

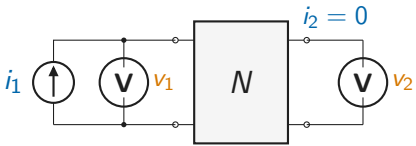
| Matrix | Inputs | 1st column | 2nd column |
|---|---|---|---|
| **R** | $i_1$, $i_2$ | port 2 open | port 1 open |
| **G** | $v_1$, $v_2$ | port 2 **shorted** | port 1 **shorted** |
| **H** | $i_1$, $v_2$ | port 2 **shorted** | port 1 open |
| **H′** | $v_1$, $i_2$ | port 2 open | port 1 **shorted** |
| **T** | $v_2$, $-i_2$ | port 2 open | port 2 **shorted** |

# Which test? Open or short

| Matrix | Inputs | 1st column | 2nd column |
|---|---|---|---|
| **R** | $i_1$, $i_2$ | port 2 open | port 1 open |
| **G** | $v_1$, $v_2$ | port 2 **shorted** | port 1 **shorted** |
| **H** | $i_1$, $v_2$ | port 2 **shorted** | port 1 open |
| **H′** | $v_1$, $i_2$ | port 2 open | port 1 **shorted** |
| **T** | $v_2$, $-i_2$ | port 2 open | port 2 **shorted** |

Zero the other input: $i = 0 \Rightarrow$ open, $v = 0 \Rightarrow$ **short**

# Measuring a two-port

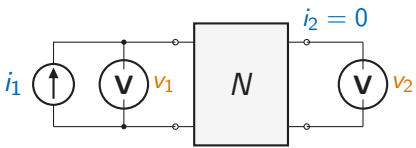



**Open-circuit test**

$$r_{11} = \frac{v_1}{i_1}, \qquad r_{21} = \frac{v_2}{i_1}$$

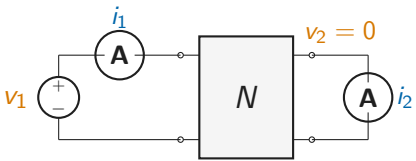
- Ideal voltmeter: an open; ideal ammeter: a **short**

# Measuring a two-port




**Open-circuit test**

$$r_{11} = \frac{v_1}{i_1}, \quad r_{21} = \frac{v_2}{i_1}$$

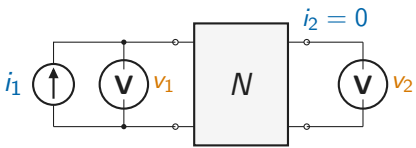



**Short-circuit test**

$$g_{11} = \frac{i_1}{v_1}, \quad g_{21} = \frac{i_2}{v_1}$$

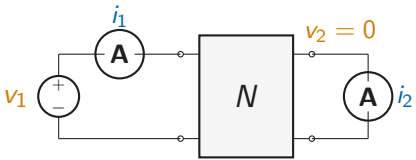
- Ideal voltmeter: an open; ideal ammeter: a **short**

# Measuring a two-port




**Open-circuit test**

$$r_{11} = \frac{v_1}{i_1}, \quad r_{21} = \frac{v_2}{i_1}$$




**Short-circuit test**

$$g_{11} = \frac{i_1}{v_1}, \quad g_{21} = \frac{i_2}{v_1}$$

- Ideal voltmeter: an open; ideal ammeter: a **short**
- One meter reading divided by the source value

# Example 14.7: R from two measurements

| | $i_1$ | $i_2$ | $v_1$ | $v_2$ |
|---|---|---|---|---|
| Exp. 1 | 1 | 1 | 5 | 5 |
| Exp. 2 | 1 | $-1$ | 3 | $-1$ |

No port open: $\mathbf{R} = ?$

# Example 14.7: R from two measurements

| | $i_1$ | $i_2$ | $v_1$ | $v_2$ |
|---|---|---|---|---|
| Exp. 1 | 1 | 1 | 5 | 5 |
| Exp. 2 | 1 | $-1$ | 3 | $-1$ |

No port open: $\mathbf{R} = ?$

$$\begin{bmatrix} 5 & 3 \\ 5 & -1 \end{bmatrix} = \mathbf{R} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

# Example 14.7: R from two measurements

| | $i_1$ | $i_2$ | $v_1$ | $v_2$ |
|---|---|---|---|---|
| Exp. 1 | 1 | 1 | 5 | 5 |
| Exp. 2 | 1 | $-1$ | 3 | $-1$ |

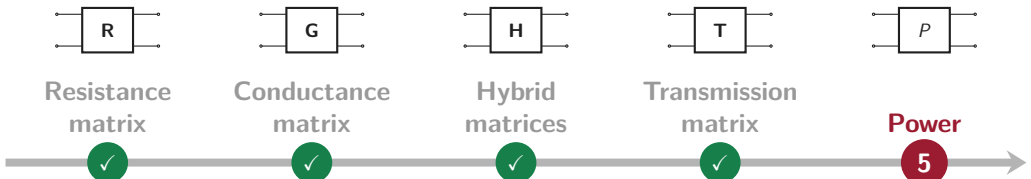
No port open: $\mathbf{R} = ?$

$$\begin{bmatrix} 5 & 3 \\ 5 & -1 \end{bmatrix} = \mathbf{R} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

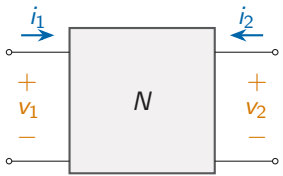
$$\mathbf{R} = \mathbf{V}\,\mathbf{I}^{-1} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Any two independent experiments

# Roadmap

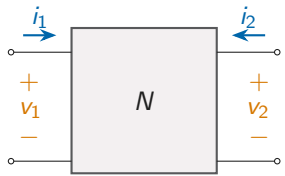

R
G
H
T
P
Resistance matrix
Conductance matrix
Hybrid matrices
Transmission matrix
Power
5
v = Ri ✓
open-circuit tests ✓
i = Gv ✓
short-circuit tests ✓
H and H′ ✓
ideal transformer ✓
$v_2$ and $-i_2$ ✓
ideal transformer ✓
$v_1 i_1 + v_2 i_2$
passive, active

# The power of a two-port




$P_N$: the sum of the powers of all elements inside $N$

# The power of a two-port

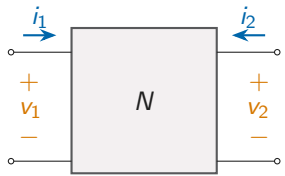



$P_N$: the sum of the powers of all elements inside $N$

$$P_N = v_1\, i_1 + v_2\, i_2$$

Proof: Tellegen's theorem (previously)

# The power of a two-port




$P_N$: the sum of the powers of all elements inside $N$

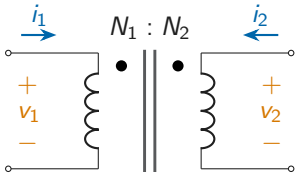
$$P_N = v_1\, i_1 + v_2\, i_2$$

Proof: Tellegen's theorem (previously)

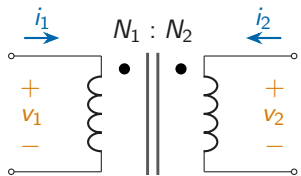
- **Passive:** $P_N \geq 0$ for all possible port variables
- **Active:** not passive

# Example 14.8: is the ideal transformer passive?

$$P_N = v_1\, i_1 + v_2\, i_2$$



# Example 14.8: is the ideal transformer passive?




$P_N = v_1\, i_1 + v_2\, i_2$

Plug $v_1 = n\, v_2$, $i_2 = -n\, i_1$:

$P_N = n\, v_2\, i_1 - n\, v_2\, i_1 = 0$

$P_N = 0$: passive

It neither absorbs nor delivers power

# Summary

| Matrix | Equation | Tests |
|---|---|---|
| resistance **R** | $[v_1 \;\; v_2]^T = \mathbf{R}\,[i_1 \;\; i_2]^T$ | ports open |
| conductance **G** | $[i_1 \;\; i_2]^T = \mathbf{G}\,[v_1 \;\; v_2]^T$ | ports shorted; $\mathbf{G} = \mathbf{R}^{-1}$ |
| hybrid **H** | $[v_1 \;\; i_2]^T = \mathbf{H}\,[i_1 \;\; v_2]^T$ | port 2 shorted, port 1 open |
| transmission **T** | $[v_1 \;\; i_1]^T = \mathbf{T}\,[v_2 \;\; -i_2]^T$ | port 1 from port 2 |

$$P_N = v_1\, i_1 + v_2\, i_2; \;\; \text{passive if } P_N \geq 0 \text{ always}$$